Math 550, S21 **Name:** Answer

Quiz 10

<u>**Instructions:**</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (8 points) Given a vector field $\mathbf{F}(x, y) = (3x^2 + y)\mathbf{i} + (\frac{z^2}{y} + x)\mathbf{j} + (2z\ln y)\mathbf{k}$. Show that $\mathbf{F}$ is a conservative vector field and then find a potential function f such that $\mathbf{F} = \nabla f$.

$$\nabla \times F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2+y & \frac{z^2}{y}+x & 2z\ln y \end{vmatrix} = \vec{i}\left[\frac{\partial}{\partial y}(2z\ln y) - \frac{\partial}{\partial z}\left(\frac{z^2}{y}+x\right)\right] - \vec{j}\left[\frac{\partial}{\partial x}(2z\ln y) - \frac{\partial}{\partial z}(3x^2+y)\right] + \vec{k}\left[\frac{\partial}{\partial x}\left(\frac{z^2}{y}+x\right) - \frac{\partial}{\partial y}(3x^2+y)\right]$$

$$= \vec{i}\left(\frac{2z}{y} - \frac{2z}{y}\right) - \vec{j}(0-0) + \vec{k}(1-1) = 0 \Rightarrow F \text{ is conservative.}$$

Next we find a potential function $f(x,y,z)$ such that $\nabla f = F$.

$$\frac{\partial f}{\partial x} = 3x^2+y \quad — ①, \quad \frac{\partial f}{\partial y} = \frac{z^2}{y}+x \quad — ②, \quad \frac{\partial f}{\partial z} = 2z\ln y \quad — ③$$

Integrating ① with respect to x, we get

$$f(x,y,z) = \int 3x^2+y \, dx = x^3+xy+g(y,z) \quad — ④$$

putting ④ into ②, we get

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^3+xy+g(y,z)) = \frac{z^2}{y}+x$$

$$x + \frac{\partial g}{\partial y}(y,z) = \frac{z^2}{y}+x$$

$$\frac{\partial g}{\partial y}(y,z) = \frac{z^2}{y}$$

$$\Rightarrow g(y,z) = \int \frac{z^2}{y} dy = z^2\ln y + h(z)$$

$$\Rightarrow f(x,y,z) = x^3+xy+z^2\ln y+h(z) \quad — ⑤$$

putting ⑤ into ③, we get

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}(x^3+xy+z^2\ln y+h(z)) = 2z\ln y$$

$$2z\ln y + h'(z) = 2z\ln y$$

$$h'(z) = 0 \Rightarrow h(z) = C \text{ where } C \text{ is any constant.}$$

Thus $f(x,y,z) = x^3+xy+z^2\ln y+C$

Problem 2. (12 points) Given a helicoid **S** parametrized by

$$\Phi(r,\theta):\ x = r\cos\theta,\ y = r\sin\theta,\ z = \theta, \quad (r,\theta)\in D,$$

where $D = \{(r,\theta) \mid 0 \le r \le 2,\ 0 \le \theta \le 2\pi\}$.

(a) Find the tangential vectors $T_r(r,\theta)$ and $T_\theta(r,\theta)$ and the normal vector $N(r,\theta)$ to this helicoid **S**.

$$T_r(r,\theta) = \frac{\partial \Phi}{\partial r}(r,\theta) = (\cos\theta, \sin\theta, 0)$$

$$T_\theta(r,\theta) = \frac{\partial \Phi}{\partial \theta}(r,\theta) = (-r\sin\theta, r\cos\theta, 1)$$

$$N(r,\theta) = T_r(r,\theta)\times T_\theta(r,\theta) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos\theta & \sin\theta & 0 \\ -r\sin\theta & r\cos\theta & 1 \end{vmatrix}$$

$$= \vec{i}\begin{vmatrix} \sin\theta & 0 \\ r\cos\theta & 1 \end{vmatrix} - \vec{j}\begin{vmatrix} \cos\theta & 0 \\ -r\sin\theta & 1 \end{vmatrix} + \vec{k}\begin{vmatrix} \cos\theta & \sin\theta \\ -r\sin\theta & r\cos\theta \end{vmatrix}$$

$$= (\sin\theta)\vec{i} - (\cos\theta)\vec{j} + r\vec{k}$$

(b) Find an equation for the plane tangent to this helicoid **S** at the point $(-1, 1, 3\pi/4)$.

$$(-1, 1, \tfrac{3\pi}{4}) \Rightarrow r\cos\theta = -1,\ r\sin\theta = 1,\ \theta = \tfrac{3\pi}{4}$$

$$\Rightarrow r = \sqrt{2},\ \theta = \tfrac{3\pi}{4}$$

Then we get $N(\sqrt{2}, \frac{3\pi}{4}) = (\sin\frac{3\pi}{4})\vec{i} - (\cos\frac{3\pi}{4})\vec{j} + \sqrt{2}\,\vec{k} = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \sqrt{2})$,

Thus the equation of tangent plane at $(-1, 1, \frac{3\pi}{4})$ is

$$\frac{\sqrt{2}}{2}(x+1) + \frac{\sqrt{2}}{2}(y-1) + \sqrt{2}\left(z - \frac{3\pi}{4}\right) = 0$$

$$\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y + \sqrt{2}z = \sqrt{2}\cdot\frac{3\pi}{4}$$

$$\underline{x + y + 2z = \frac{3\pi}{2}}$$

(c) Compute the integral $\iint_\Phi \sqrt{x^2+y^2}\, dS := \iint_D \sqrt{x^2+y^2}\|N(r,\theta)\|\, drd\theta$.

$$\iint_\Phi \sqrt{x^2+y^2}\,dS = \iint_D \sqrt{x^2+y^2}\,\|N(r,\theta)\|\,drd\theta$$

$$= \int_0^{2\pi}\int_0^2 r\cdot\sqrt{(\sin\theta)^2 + (-\cos\theta)^2 + r^2}\,drd\theta$$

$$= \int_0^{2\pi}\int_0^2 r\sqrt{r^2+1}\,drd\theta$$

$$= \int_0^{2\pi}\left[\frac{1}{3}(r^2+1)^{3/2}\right]_{r=0}^{r=2} d\theta = \int_0^{2\pi}\frac{1}{3}(5^{3/2}-1)\,d\theta$$

$$= \frac{2\pi}{3}(5^{3/2}-1)$$